

Question 1. (12 Marks) Use a SEPARATE Writing Booklet.

Marks

(a) Simplify $\frac{8x^3 - 27}{2x^2 - x - 3}$.

2

(b) Solve $\tan 2\theta = \frac{1}{\sqrt{3}}$, $0^\circ \leq \theta \leq 360^\circ$.

2

(c) Find the point P, which divides the interval joining $A(1, -2)$ and $B(-4, 0)$ externally in the ratio 3:5.

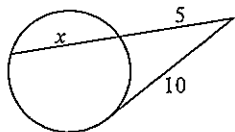
2

(d) State the locus of a point which moves so that it is always 2 units from the line $y = -1$.

1

(e) Find x in the diagram below.

1



(f) Find the values of A , B and C if $2x^2 + 8x - 6 \equiv A(x+1)^2 + Bx + C$.

2

(g) If α and β are roots of the equation $2x^2 + 3x - 4 = 0$, find the value of:

(i) $\alpha + \beta$ and $\alpha\beta$

1

(ii) $\alpha^2 + \alpha\beta + \beta^2$

1

Question 2. (12 Marks) Use a SEPARATE Writing Booklet.

Marks

(a) Find the co-ordinates of the focus of the parabola with equation $y^2 = -12x$.

1

(b) Find the equation of the directrix of the parabola $y = x^2 + 3x$.

2

(c) Find and describe the locus of a point which moves so that it is equidistant from the lines $y = 5$ and $3x - 4y + 6 = 0$.

3

(d) Solve $\frac{2x+3}{x-4} > 1$.

3

(e) Prove that $\frac{\tan\theta + \cot\theta}{\sec\theta \operatorname{cosec}\theta} = 1$.

3

Question 3. (12 Marks) Use a SEPARATE Writing Booklet.

Marks

- (a) Find the values of k for which the equation $x^2 + (k+3)x + k(k+3) = 0$ has 2 real and distinct roots. 3

- (b) (i) Sketch $y = |2x-5|$ showing x and y intercepts. 1

- (ii) Graphically or otherwise, solve $|2x-5| = x-1$. 2

- (iii) Find all values of x such that $|2x-5| < x-1$. 1

- (c) Graph the region $y \leq \sqrt{4-x^2}$. 2

- (d) Solve $2^{2x+3} + 6 \cdot 2^x - 5 = 0$. 3

Question 4. (12 Marks) Use a SEPARATE Writing Booklet.

Marks

- (a) (i) Sketch the curves $y = (x-1)(x+2)$ and $y = (x-1)(x-3)$ and hence find their point of intersection. 1

- (ii) Find the acute angle between the tangents to each curve at this point. 3

- (b) (i) Prove that $x^2 + x + 1 = 0$ has no real roots. 2

- (ii) Hence show that the curve with equation $y = 2x^3 + 3x^2 + 6x$ has no stationary points. 2

- (iii) For what values of x is this curve concave up? 2

- (iv) Sketch $y = 2x^3 + 3x^2 + 6x$. 2

Question 5. (12 marks) Use a SEPARATE Writing Booklet.

Marks

- (a) The variable point $\left(\frac{t}{4}, 6t^2\right)$ lies on a parabola.

Find the Cartesian equation for the parabola.

1

- (b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

- (i) PQ is a focal chord. Show that $pq = -1$.

2

- (ii) Show that the coordinates of the midpoint, M , of the chord PQ are $\left[a(p+q), \frac{a}{2}(p^2+q^2)\right]$.

1

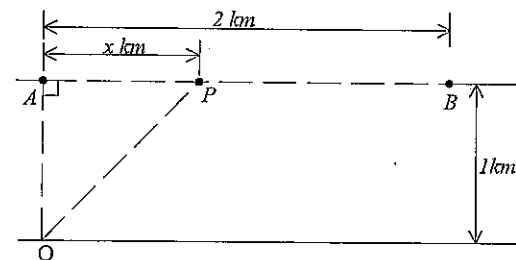
- (iii) Find the equation of the locus of M and describe the locus of M geometrically.

3

Question 5 continued on page 6.

Question 5. (continued)

(c)



The diagram shows a straight section of river, one kilometre wide. Alice is at a point O on one bank and she wishes to reach a point B on the opposite bank. The point A is directly opposite O and the distance from A to B is two kilometres.

Alice can row at 6 km/h and jog at 10 km/h. She intends to row in a straight line to a point P on the opposite bank and then jog directly from P to B .

Let the distance AP be x kilometres.

- (i) Show that the time T , in hours, that Alice takes to reach B is given by

1

$$T = \frac{\sqrt{x^2 + 1}}{6} + \frac{2-x}{10}.$$

- (ii) Show that if Alice wishes to minimise the time taken to complete the journey then she should row to a point P , $\frac{3}{4}$ kilometre from A .

4

Question 6 is on the next page.

Question 6. (12 marks) Use a **SEPARATE** Writing Booklet.

Marks

(a) (i) Show that $\frac{2x^2}{x^2-1} = 2 + \frac{2}{(x-1)(x+1)}$. 1

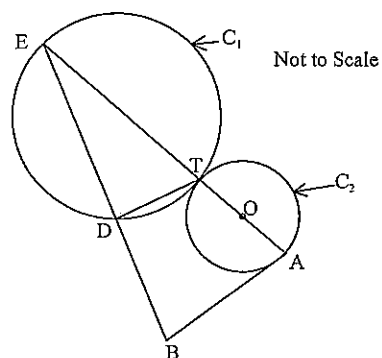
(ii) Show that $f(x) = \frac{2x^2}{x^2-1}$ is an even function. 1

(iii) Find the equations of any vertical or horizontal asymptotes. Justify your answers. 2

(iv) Sketch $y = f(x)$. 2

(v) What is the range of the function? 1

(b)



Two circles C_1 and C_2 touch at T . The line AE passes through O , the centre of C_2 , and through T .

The point A lies on C_2 and E lies on C_1 .

The line AB is a tangent to C_2 at A . D lies on C_1 and BE passes through D .

The radius of C_1 is R and the radius of C_2 is r .

(i) Prove that $\angle EDT = 90^\circ$. 2

(ii) If $DE = 2r$, find an expression for the length of EB in terms of r and R . 3

End of Paper

Extension 1 Yearly 2011.

Question 1.

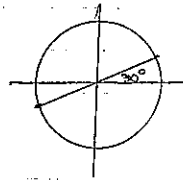
$$a) \frac{8x^3 - 27}{2x^2 - x - 3} = \frac{(2x-3)(4x^2+6x+9)}{(2x-3)(x+1)} \leftarrow (1)$$

$$= \frac{4x^2+6x+9}{x+1} \quad (1)$$

$$b) \tan 2\theta = \frac{1}{\sqrt{3}} \quad 0^\circ \leq \theta \leq 360^\circ$$

$$2\theta = 30^\circ, 210^\circ, 390^\circ, 570^\circ \quad 0^\circ \leq 2\theta \leq 720^\circ$$

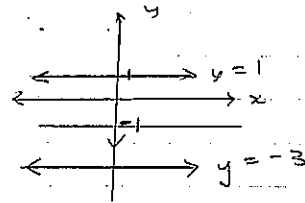
$$\theta = 15^\circ, 105^\circ, 195^\circ, 285^\circ$$



$$c) \left(\frac{5 \times 1 + (-3) \times (-4)}{-3 + 5}, \frac{5 \times -2 + (-3) \times 0}{-3 + 5} \right)$$

$$\left(\frac{17}{2}, -5 \right) \quad (1)$$

$$d) y = 1 \text{ or } y = -3$$



R/W (1)

$$e) (x+5) \times 5 = 10^2$$

$$5x + 25 = 100$$

$$5x = 75$$

$$x = 15$$

R/W (1)

$$f) 2x^2 + 8x - 6 \equiv Ax^2 + 2Ax + A + Bx + C$$

$$\therefore A = 2 //$$

$$2A + B = 8$$

$$B = 8 - 2 \times 2 = 4 //$$

$$A + C = -6$$

$$C = -6 - 2 = -8 //$$

(1) 2 correct answers

(1) 3rd correct answer

$$g) i) \alpha + \beta = -3/2 //$$

$$ii) = (\alpha + \beta)^2 - \alpha\beta$$

$$= (-3/2)^2 - (-2)$$

R/W (1)

Question 2.

$$a) y^2 = -12x$$

$$4a = 12$$

$$\therefore a = 3$$

Focus is $(-3, 0) //$

(1) R/W

$$b) y = x^2 + 3x$$

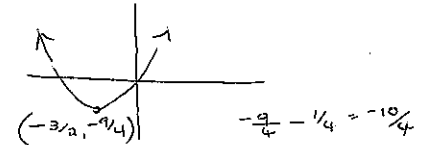
$$= (x + 3/2)^2 - 9/4$$

$$\therefore (x + 3/2)^2 = y + 9/4 \quad (1)$$

$$4a = 1$$

$$a = 1/4$$

$\therefore$ Directrix is $y = -5/4 //$ (1)



c) Let the point be $P(x, y)$

then

$$\frac{|3x - 4y + 6|}{\sqrt{3^2 + 4^2}} = \frac{|y - 5|}{\sqrt{1^2}} \quad (1)$$

$$|3x - 4y + 6| = 5|y - 5|$$

$$\therefore \text{Either } 3x - 4y + 6 = 5y - 25 \text{ or } 3x - 4y + 6 = 25 - 5y$$

$$(1) \underline{\text{Lines}} \quad 3x - 9y + 31 = 0 \text{ or } 3x + y - 19 = 0 \quad (1)$$

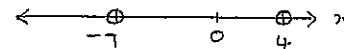
$$d) \frac{2x+3}{x-4} > 1$$

$$\text{CP: } x = 4$$

$$\text{Consider } \frac{2x+3}{x-4} = 1$$

$$\frac{2x+3}{x-4} = x-4$$

$$\text{CP: } x = -7$$



$$\text{Test } x = -8 \quad \frac{2 \times -8 + 3}{-8 - 4} = \frac{-13}{-12} > 1 \therefore \checkmark$$

$$x = 0 \quad \frac{2 \times 0 + 3}{0 - 4} = -3/4 < 1 \therefore \times$$

$$x = 5 \quad \frac{2 \times 5 + 3}{5 - 4} = 13 > 1 \therefore \checkmark$$

(1)

$$\therefore x < -7 \text{ or } x > 4$$

$$e) \text{LHS} = \frac{\tan \theta + \cot \theta}{\sec \theta \csc \theta}$$

$$= \frac{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}}{\frac{1}{\cos \theta} \times \frac{1}{\sin \theta}} \times \cos \theta \sin \theta$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{1} = 1$$

Question 3

a) $x^2 + (k+3)x + k(k+3) = 0$

For real and distinct roots $\Delta > 0$

$(k+3)^2 - 4 \cdot 1 \cdot k(k+3) > 0$

① calculating $k^2 + 6k + 9 - 4k^2 - 12k > 0$

Δ correctly $3k^2 + 6k - 9 < 0$

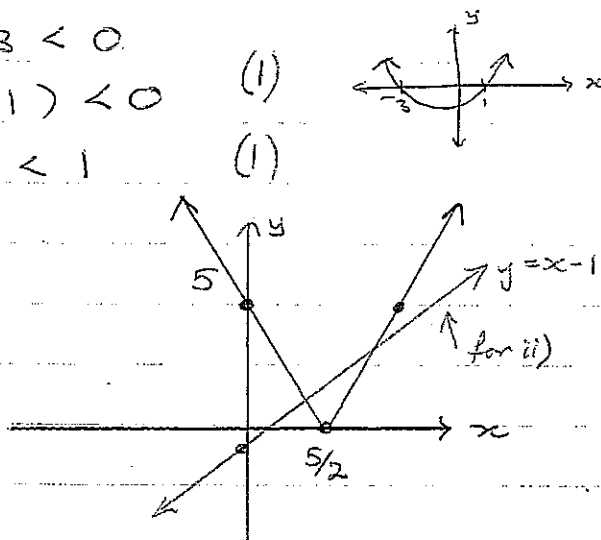
and stating $\Delta > 0$ $k^2 + 2k - 3 < 0$

① Setting up a $(k+3)(k-1) < 0$ (1)

correct inequality $-3 < k < 1$ (1)

① Solⁿ

b) i) $y = |2x - 5|$



(1)
r/w

ii) $2x - 5 = x - 1$ or $-(2x - 5) = x - 1$

$x = 4$ // (1)

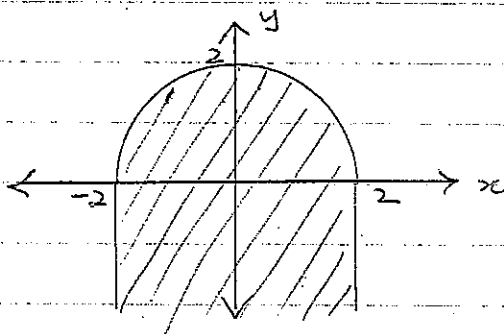
$2x - 5 = 1 - x$

$3x = 6$

$x = 2$ // (1)

iii) $2 < x < 4$ (1) r/w

c)



(1) correct shape and intercept

(1) correct shading and boundary lines

d) $2^{2x+3} + 6 \cdot 2^x - 5 = 0$

$2^3 \cdot 2^{2x} + 6 \cdot 2^x - 5 = 0$

Let $a = 2^x$

$8a^2 + 6a - 5 = 0$

$(4a + 5)(2a - 1) = 0$

$a = -5/4$ or $a = 1/2$

$\therefore 2^x = -5/4$ or $2^x = 1/2$

no solution

$x = -1$

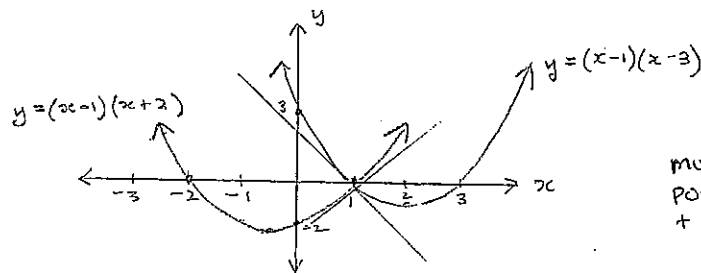
① For creation of quadratic

① For solⁿ and resubst^d

① disqualification of 1 solⁿ

Question 4.

a) i)



Point of intersection (1,0)

must state
point of intersection
+ include sketch
(1)

ii) $y = (x-1)(x+2)$
 $y = x^2 + x - 2$

$$\frac{dy}{dx} = 2x + 1$$

At (1,0) $\frac{dy}{dx} = 2 \times 1 + 1 = 3$

$y = (x-1)(x-3)$
 $y = x^2 - x - 3x + 3$
 $y = x^2 - 4x + 3$

$$\frac{dy}{dx} = 2x - 4$$

At (1,0) $\frac{dy}{dx} = 2 \times 1 - 4 = -2$

Acute Angle between tangents, α , is:

m - all correct
m - correct derivs + grad
+ incorrect use of formula
or
incorrect derivs with
correct grad
+ correct use of formula
or
correct derivs, incorrect
grads + correct use of formula.

$$\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad \text{where } m_1 = 3 \text{ and } m_2 = -2$$

$$= \left| \frac{3 - (-2)}{1 + 3 \times (-2)} \right|$$

$$= \left| \frac{5}{-5} \right|$$

$$= 1$$

$$\therefore \alpha = 45^\circ //$$

m - correct derivs.
or correct use of formula (derivs + grad incorrect)

b) i) $x^2 + x + 1 = 0$

$$\Delta = 1^2 - 4 \cdot 1 \cdot 1 = -3$$

(1)

Since $\Delta < 0$, $x^2 + x + 1 = 0$ has no real roots. (1)

ii) For stationary points $\frac{dy}{dx} = 0$

$$\frac{dy}{dx} = 6x^2 + 6x + 6$$

$$= 6(x^2 + x + 1)$$

Since $x^2 + x + 1$ has no real roots, there are no stat pts

Question 4 continued.

iii) For concave up $\frac{d^2y}{dx^2} > 0$

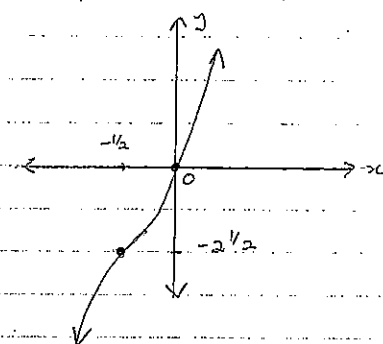
$$\frac{d^2y}{dx^2} = 12x + 6$$

$$12x + 6 > 0 \quad (1)$$

$$x > -\frac{6}{12}$$

$$x > -\frac{1}{2} //$$

iv)



When $x = -\frac{1}{2}$,

$$y = 2\left(-\frac{1}{2}\right)^3 + 3\left(-\frac{1}{2}\right)^2 + 6\left(-\frac{1}{2}\right)$$

$$= -\frac{1}{4} + \frac{3}{4} - 3$$

$$= -2\frac{1}{2}$$

1 - Correctly placed cubic
passing through origin.

1 - labelling point of inflexion.

Question 5

$$\begin{aligned} \text{a) } (t/4, bt^2) \quad x &= t/4 \quad y = bt^2 \\ \therefore t &= 4x \\ y &= b(4x)^2 \\ &= 16bx^2 \quad (1) \end{aligned}$$

$$\text{b) } x^2 = 4ay$$

$$\begin{aligned} \text{Gradient of PQ is } \frac{aq^2 - ap^2}{2aq - 2ap} &= \\ &= \frac{a(q-p)(q+p)}{2a(q-p)} \\ &= \frac{q+p}{2} \end{aligned}$$

$$\therefore \text{Equation of PQ is } y - ap^2 = \frac{q+p}{2}(x - 2ap) \quad (1)$$

Since PQ is a focal chord, $(0, a)$ lies on this line

$$\begin{aligned} a - ap^2 &= \frac{q+p}{2}(0 - 2ap) \\ 2a - 2ap^2 &= -2apq - 2ap^2 \\ 2a &= -2apq \\ 1 &= -pq \\ pq &= -1 \end{aligned} \quad (1)$$

$$\begin{aligned} \text{ii) Midpoint of PQ is } \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right) & \quad (1) \\ &= \left(a(p+q), \frac{a(p^2 + q^2)}{2} \right) \end{aligned}$$

$$\begin{aligned} \text{iii) M: } x &= a(p+q) \quad y = \frac{a}{2}(p^2 + q^2) \\ \Rightarrow p+q &= \frac{x}{a} \quad = \frac{a}{2}[(p+q)^2 - 2pq] \quad (1) \end{aligned}$$

$$\text{But } p+q = x/a \text{ and } pq = -1$$

$$\therefore y = \frac{a}{2} \left[\left(\frac{x}{a} \right)^2 - 2(-1) \right]$$

5biii) continued

$$\begin{aligned} y - a &= x^2/2a \\ x^2 &= 2a(y - a) \end{aligned}$$

Hence the locus of M is a parabola with Vertex $(0, a)$ and focal length $\frac{a}{2}$ } (1)

$$\text{5c) i) } OP = \sqrt{x^2 + 1} \quad \text{By Pythagoras}$$

$$\therefore \text{Rowing time} = \frac{\sqrt{x^2 + 1}}{6} \quad S = D/T, T = D/S$$

$$\text{Jogging time} = \frac{2-x}{10} \quad (1)$$

$$\therefore \text{Time taken to reach B is } T = \frac{\sqrt{x^2 + 1}}{6} + \frac{2-x}{10}$$

ii) To minimise journey, let $dT/dx = 0$ and $d^2T/dx^2 > 0$ (or change in sign of dT/dx from - to +)

$$\begin{aligned} \frac{dT}{dx} &= \frac{1}{6} \cdot \frac{1}{2}(x^2 + 1)^{-1/2} \cdot 2x - \frac{1}{10} \\ &= \frac{x}{6\sqrt{x^2 + 1}} - \frac{1}{10} \\ &= 0 \quad \text{when} \quad \frac{x}{6\sqrt{x^2 + 1}} = \frac{1}{10} \end{aligned} \quad (1)$$

$$10x = 6\sqrt{x^2 + 1}$$

$$100x^2 = 36(x^2 + 1)$$

$$64x^2 = 36$$

$$x^2 = 36/64$$

$$x = 6/8 = 3/4$$

(1) $x > 0$ (1)
It's a distance

Check for minimum time

x	$1/2$	$3/4$	1
dT/dx	$\frac{1}{6\sqrt{5/4}} - \frac{1}{10}$	0	$\frac{1}{6\sqrt{2}} - \frac{1}{10}$

$\therefore dT/dx$ changes in sign from negative to positive either side

Question 6

$$\begin{aligned}
 \text{a) i) RHS} &= 2 + \frac{2}{(x-1)(x+1)} \\
 &= \frac{2(x-1)(x+1) + 2}{(x-1)(x+1)} \\
 &= \frac{2(x^2-1) + 2}{(x-1)(x+1)} \\
 &= \frac{2x^2 - 2 + 2}{(x-1)(x+1)} \\
 &= \frac{2x^2}{(x-1)(x+1)} \\
 &= \text{LHS}
 \end{aligned}$$

$$\therefore \frac{2x^2}{x^2-1} = 2 + \frac{2}{(x-1)(x+1)} \quad \textcircled{1} \text{ r/w}$$

OR

$$\begin{array}{r}
 x^2-1 \quad \left. \begin{array}{l} 2x^2 \\ -(2x^2-2) \end{array} \right\} \\
 \hline
 2
 \end{array}$$

$$\begin{aligned}
 \therefore \frac{2x^2}{x^2-1} &= 2 + \frac{2}{x^2-1} \\
 &= 2 + \frac{2}{(x-1)(x+1)}
 \end{aligned}$$

$$\text{ii) } f(x) = \frac{2x^2}{x^2-1}$$

$$f(-x) = \frac{2(-x)^2}{(-x)^2-1} = \frac{2x^2}{x^2-1} \quad \textcircled{1} \text{ r/w}$$

Since $f(-x) = f(x)$, $f(x)$ is an even function

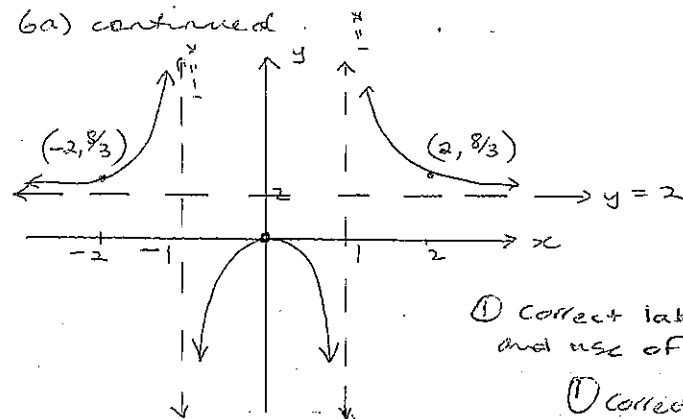
iii) Vertical Asymptotes: $x = \pm 1$ since $x^2-1 \neq 0$

Horizontal Asymptote $y = 2$ since

$$\frac{2}{(x-1)(x+1)} \rightarrow 0 \text{ as } x \rightarrow \pm\infty \text{ so } 2 + \frac{2}{(x-1)(x+1)} \rightarrow 2$$

Question 6a) continued

iv)



① Correct labelling and use of asymptotes

① Correct graph

Correctly positioned

v) Range: $y \leq 0$; $y > 2$ ① r/w

6.b) i) When 2 circles touch, the line of centres passes through the point of contact. ①
 $\therefore$ ET is a diameter of C_1
 $\therefore \angle EDT = 90^\circ$ Angle in a semi-circle is a right angle ①

ii) In $\triangle DET$ and $\triangle AEB$
 $\angle E$ is common

If there is a justification of why $\triangle DET$ is right angled

① Similarity $\angle EDT = 90^\circ$ from (i)

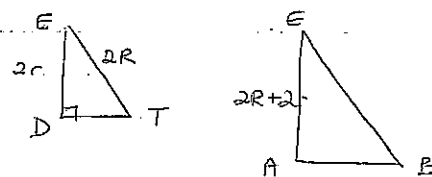
Proof $\angle EAB = 90^\circ$ (angle between a tangent and radius at point of contact is 90°)

① Corresponding Sides Comparison

$$\angle EDT = \angle EAB$$

① Expression $\therefore \triangle DET \parallel \triangle AEB$ (equiangular)

Here:



(Note $ED = 2r$ is given)

$$\begin{aligned}
 \frac{EB}{ET} &= \frac{EA}{ED} \quad (\text{ratio of corresponding sides in similar } \Delta\text{'s}) \\
 EB &= 2(R+r)
 \end{aligned}$$